# Impulse changes momentum

**antocciphysics**

UNIT 3 / DAY 3 / HONORS

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect force over time to momentum change.

**SUCCESS CRITERIA - I can...**

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

An ideal circular orbit has GM=225 $m^3/s^2$ and r=9 m. Find speed, inward acceleration and period; explain how velocity changes.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; $F_{avg}$ is average net force. Signed force-time area gives impulse; i = initial, f = final.

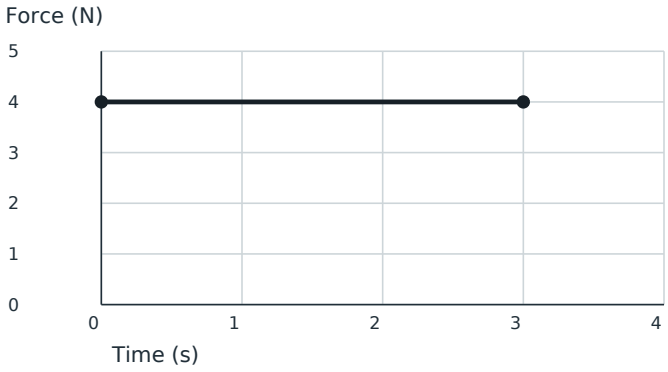
**Worked reasoning:** A constant 6 N force acts right for 2 s. Impulse is 12 N·s right, so momentum increases by 12 kg·m/s.

**Your turn:** Draw the force-time rectangle and shade its area. Explain why the two impulse units are equivalent.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

Read the rectangular force-time graph. Find the area and state what it means.




**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A .50 kg ball changes velocity from +8 to -6 m/s in .020 s. Find impulse and average net force.

**Marzano check - circle:** ① ② ③

# Impulse changes momentum

**antocciphysics**



Continue down each column.

### 05 Analyze the reasoning

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

At what altitude h/R is a spherical planet’s field 1/16 of its surface value? Repeat for 1/36.

**Marzano check - circle:** 1 2 3

### 07 Exit check

A triangular force pulse has base .20 s and peak 30 N. Find impulse and velocity change for a .60 kg object.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A 2 kg cart changes velocity from +3 to -2 m/s in 0.1 s. Find impulse and average force. Sketch a force-time rectangle with the correct signed area.

**Marzano check - circle:** 1 2 3